FORSYS
Forward Thinking

Case Study

Automated Quality Assurance Platform & Streamlined E2E QA Process

Insight

The client is a prominent e-learning company producing learning content to fit different clients' training needs. Once developed, the courses can be accessed via the web and/or a wide range of mobile devices. Owing to the unavailability of trained testers and inadequate training processes, the customer's QA process lacked structure and needed Web, Mobile and API testing support for numerous projects.

Forsys provided E2E manual and automated quality assurance services including Web, Mobile and API testing for numerous projects using a wide range of frameworks, tools and devices, leading to curtailed test cycle time, and improved test coverage platform quality.

Industry	Revenue	Employees	Headquarters
E-Learning	$400+ M	15000+	Mountain View, California

Business Challenge

Managing end-to-end manual and automated quality assurance for different projects proved to be a challenge for the customer. Reason? Each project and others needed thousands of test cases.

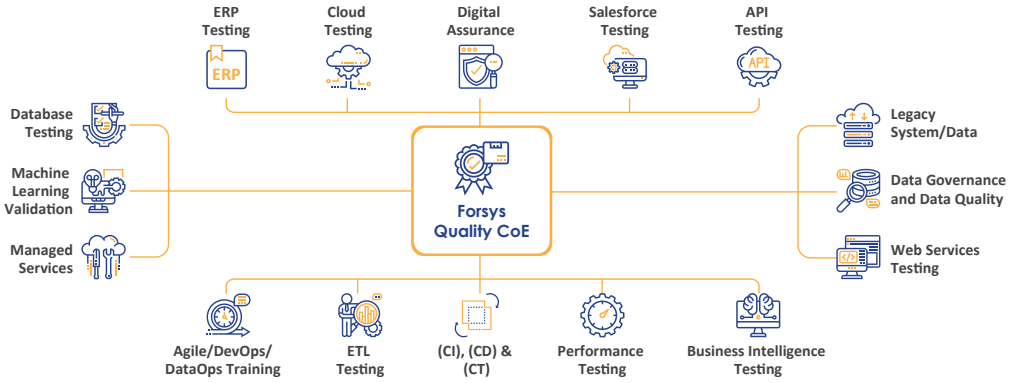
While the online learning client had testers with a broad range of skill sets but preferred a solution that did not require deep technical experience so that testers could focus on providing subject matter expertise rather than spending time learning programming languages to conduct automated testing.

Transformation

From 2018 onwards, Forsys has been delivering end-to-end manual and automated quality assurance services to the client including Web, Mobile and API testing for numerous projects using a wide range of frameworks, tools and devices. The project scope included automating the client's quality assurance platform, identifying the challenges, creating test case templates and structures, and bringing improvement in test case maintenance.

Our solution approach focused on:

- Leveraged Page Object Model (POM) framework for E2E testing.
- Developed the initial templates and structure in the form of a POC which was further built upon to implement tests in the POM framework by reducing code duplication.
- Implemented test strategy and test planning documentation while ensuring adherence to ISTQB standards.
- Improved test case maintenance with reusable utils and functions.
- Constructed a collection of tests as a regression suite using Postman for API testing.
- Implemented JIRA tracking to improve the end-to-end QA process.
- Maintained constant coordination with the development team to identify and execute scenarios with maximum test coverage.
- Embedded continuous testing within deployment pipelines for continuous development and continuous delivery.



Impact

With Forsys, the client was able to realize their testing vision and automation goals with confidence.

Some of the benefits the customer witnessed include:

- Streamlined requirement enhancement documentation to ensure maximum test coverage and accurate bug reporting.
- Streamlined test data gathering and process defined.
- Quick scaling up in Puppeteer to deliver POM framework using JavaScript.
- Users had a great experience on its online platforms no matter what device they used.
- Reduced test cycle time and improved test coverage.
- Improved time-to-market and platform quality.

About the Client

The client is a leading US-based massive open online course provider. Founded in 2012 by Stanford professors, the e-learning provider offers a range of learning opportunities—from hands-on projects and courses to job-ready certificates, career credentials, and degree programs – and partners with more than 200 leading universities and companies to bring flexible, affordable, job-relevant online learning to individuals and organizations worldwide.

75+ million learners, 100+ Fortune 500 companies, and more than 6,400 campuses, businesses, and governments access the client's world-class learning—anytime, anywhere.

Solution Components

Manual

- Functional Testing
- Performance and Load Testing
- API Testing
- User Interface Testing
- End-to-End Testing
- Integration Testing
- Compatibility Testing
- Smoke Testing
- Regression Testing
- Accessibility Testing

Automation

- Python
- Selenium + JAVA
- TestNG
- Protractor
- CI/CD - Jenkins
- Puppeteer
- Testim
- Appium